

AI Career Guidance & Job Matching Assistant

1. Project Overview

The **AI Career Guidance & Job Matching Assistant** is an AI-powered career support system developed using **Dify**. It analyzes a student's education, skills, CGPA, backlogs, and career requirements to provide personalized career assistance.

2. Objectives

- Help students identify suitable career opportunities.
- Check placement eligibility.
- Identify skill gaps.
- Prepare students for interviews.
- Create personalized learning plans.
- Provide general career guidance.

3. Key Features

1. **Job Matching** – Matches student skills with required job skills and calculates a skill-match percentage.
2. **Placement Eligibility** – Analyzes education, CGPA, backlogs, skills, and job requirements.
3. **Interview Preparation** – Provides technical topics, interview questions, practical questions, and preparation tips.
4. **Skill Gap Analysis** – Identifies existing and missing skills.
5. **Learning Plan** – Creates a structured learning plan based on the career goal.
6. **Career Guidance** – Suggests suitable career paths and next steps.

4. Technologies Used

- Dify
- Large Language Model (LLM)
- Groq
- Knowledge Base
- Knowledge Retrieval
- Question Classifier

- Parameter Extractor
- Python
- HTML, CSS, JavaScript knowledge for career matching examples

5. How the System Works

The student provides:

- Name
- Education
- Skills
- CGPA
- Backlogs
- Career query

The system analyzes the career query and classifies it into one of six categories. Based on the category, the appropriate AI workflow provides the required result.

6. Knowledge Base

A **Career Job Requirements** knowledge base is used to store job roles and their required technical skills.

Example:

Job Role	Required Skills
Frontend Developer Intern	HTML, CSS, JavaScript, Bootstrap, Git
Backend Developer Intern	Python, SQL, REST API, Git
Data Analyst Intern	Python, SQL, Excel, Power BI, Pandas
AI/ML Intern	Python, NumPy, Pandas, Machine Learning, Scikit-learn
Full Stack Developer Intern	HTML, CSS, JavaScript, React, Node.js, SQL, Git

7. Workflow

Student Input → Career Analysis → Parameter Extraction → Question Classification → Appropriate Career Assistance → Final Result

For job matching:

Career Query → Knowledge Retrieval → Required Skill Extraction → Skill Match Calculation → Job Matching Result

8. Expected Outcome

The system helps students understand their career options, evaluate their current skills, identify areas for improvement, and prepare for internships and placements.

9. Future Improvements

- Add more job roles and companies.
- Add real-time job opportunities.
- Add resume analysis.
- Add personalized course recommendations.
- Add interview chatbot functionality.
- Add real-time job application tracking.

10. Conclusion

The **AI Career Guidance & Job Matching Assistant** demonstrates how AI and workflow automation can be used to provide personalized career support for students. The system combines career analysis, knowledge retrieval, skill matching, eligibility checking, interview preparation, and learning recommendations in one platform.